

Khalil Lagha



M2 student in Intelligent Systems for Automotive & Aerospace (SIAA) — embedded systems, control & power electronics · Université Paris-Saclay · Work-study via CFA EVE

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SUMMARY

M2 student in Intelligent Systems for Automotive & Aerospace (SIAA, Université Paris-Saclay, **work-study via CFA EVE**), **ranked 2nd of the M1 EEA cohort (15.43/20)**. Specialized in **control and embedded systems**: field-oriented control (**FOC**), **PID** controllers, observers, **Kalman filter**, **Matlab/Simulink** validation (specification → implementation → test) then real-hardware testing — across **five internships** from the lab (GIPSA-lab) to industrial sites. Target fields: **EV / powertrain, embedded systems, applied AI**. Seeking: **work-study contract (alternance) from September 2026 — 2 days company / 3 days university**.

EXPERIENCE

Research Intern — GIPSA-lab, Grenoble

(04/2026 - 07/2026, 3.5 months)

- Modeled a **10-cell PEMFC fuel-cell stack** in Matlab/Simulink — **specification** → **implementation** → **test** approach; achieved **6%** dynamic error and **1%** static error against real measurements
- Tuned a **DC-DC boost converter PI regulation loop** to optimize efficiency; validated first in simulation, then on a real test bench (tests, measurements, documentation)

Intern — SLB (Schlumberger)

(07/2025 - 07/2025, 1 month)

- Authored technical summaries in **English** (directional drilling) in an international industrial environment

Power Electronics & Automation Intern — EURL Lagha

(01/2024 - 01/2024, 15 days · 09/2024 - 09/2024, 1 month)

- Contributed to the electronics of real industrial equipment: a **DC-DC buck converter** (50 V → adjustable 0–48 V, **25 A**) and **4 transformer-rectifiers** (100 V DC / 80 A ≈ 8 kW, ripple < 5%)
- Co-designed the supervision HMI (**Siemens TIA Portal**, Ladder PLC); configured industrial control applications
- Analyzed a customer specification and sized the corresponding transformer; produced the electrical schematics in **EPLAN** — direct interface with manufacturing

Grid Intern — Sonelgaz

(03/2024 - 03/2024, 15 days, observation)

- Analyzed the full generation → distribution chain (**EHV 400/220 kV** → **HV 60 kV** → **MV 30/10 kV** → **LV 400/230 V**) and substation architecture; surveyed an MV/LV substation during a technical site visit

EDUCATION

- M2 SIAA — Intelligent Systems for Automotive & Aerospace**, Université Paris-Saclay (Évry) · CFA EVE — from September 2026, work-study
- M1 EEA (EECS)**, Université Grenoble Alpes (2025–2026) — **ranked 2nd of the cohort, annual average 15.43/20**
- Engineering cycle in Electrical Engineering**, École Nationale Polytechnique, Algiers (2021–2025) — 4 of 5 years completed; prep classes: among the top-ranked
- Baccalauréat in Mathematics** (2021) — highest honors (16.93/20)

PROJECTS

- Field-oriented control (FOC) of an induction motor** (Matlab/Simulink) — implemented Park transforms, PID controllers, observers and a **Kalman filter**; tuned the **cascaded current- and speed-regulation loops** — the core of electric-vehicle traction
- PWM generator with adjustable frequency and duty cycle** (555 timer + D flip-flops) — designed, simulated in **Proteus**, then **built and fine-tuned on real hardware** — a complete electronics-diagnostics workflow
- Neural network from scratch (MLP)** (Matlab, open source) — hand-coded a multilayer perceptron and its **backpropagation** without any toolbox; published under the MIT license — github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab
- Electrical machine sizing** (Python, team) — built a **Python (OOP, PyQt)** application: from specification to automatic parameter generation; finite-element validation (**FEMM**)
- Also: cascade control of a DC motor (three-phase thyristor bridge, Simulink); Siemens Step7 automation (HMI + Ladder); Arduino line-follower robot (Poly Maze competition); power grid analysis (load flow, 3–118 buses, Matlab)

SKILLS

- Control engineering**: PID, LQR, observers, Kalman filter; field-oriented control (FOC); cascade control; regulation-loop tuning; system identification
- Embedded systems & programming**: C/C++, Python (OOP, PyQt), Arduino (IDE + firmware); hands-on electronics, from schematic to real circuit (Proteus, LTspice)
- Industrial automation**: Siemens PLC (TIA Portal, Step7 — Ladder, SCL, FBD, SFC); HMI/SCADA; Grafcet
- AI**: neural networks (backpropagation), genetic algorithms, data fusion, fuzzy logic
- Power electronics**: DC-DC conversion (buck, boost), AC-DC, DC-AC, AC-AC; PWM; power electronics for machine control
- Software**: Matlab/Simulink · Python (OOP, PyQt) · C/C++ · Arduino · Proteus · LTspice · Siemens TIA Portal · Step7 (Ladder/SCL) · EPLAN · AutoCAD Electrical · FEMM · SolidWorks · Git/GitHub

SOFT SKILLS

Fast learner • Comfortable with tools/software • Problem solving • Rigorous • Organized — IEEE Student Branch ENP • VIC (ENP)

CERTIFICATIONS

- SLB — NEST & SIPP Level 2 (07/2025) — HSE, industrial-operations safety

LANGUAGES

French — C1 • English — C2